Loan approval assignment

ITC 3154 Cognitive Computing
Assignment 1 Fall 2026
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Code repository: Pending - repository has not been created

Tasks 1 and 2

For this assignment I looked at the loan approval dataset to see which columns seem related to getting a loan approved. I used Python to open the file, check the data and make the graphs.

I loaded loan_approval_dataset.csv using pandas. The file separates the values with semicolons, so I used sep=";". I also removed spaces around the column names and text values. I checked for missing values and repeated rows. There were none, so I did not remove any rows.

Check	Result
Applications	4269
Original columns	13
Missing cells	0
Duplicate rows	0
Unique loan identifiers	4269
Approved	2656 (62.22%)
Rejected	1613 (37.78%)

I used df.head(20) to get the first 20 rows and printed them with all the columns showing. The picture on the next page shows these rows. I split it into two parts so it is easier to read. The loan_id column is in both parts to show which values belong to the same person.

Task 2 First 20 rows

First 20 rows after trimming whitespace - all 13 original columns

loan id	no of_dependents	education	self employed	income annum	loan amount	loan term
1	2	Graduate	No	9600000	29900000	12
2	0	Not Graduate	Yes	4100000	12200000	8
3	3	Graduate	No	9100000	29700000	20
4	3	Graduate	No	8200000	30700000	8
5	5	Not Graduate	Yes	9800000	24200000	20
6	0	Graduate	Yes	4800000	13500000	10
7	5	Graduate	No	8700000	33000000	4
8	2	Graduate	Yes	5700000	15000000	20
9	0	Graduate	Yes	800000	2200000	20
10	5	Not Graduate	No	1100000	4300000	10
11	4	Graduate	Yes	2900000	11200000	2
12	2	Not Graduate	Yes	6700000	22700000	18
13	3	Not Graduate	Yes	5000000	11600000	16
14	2	Graduate	Yes	9100000	31500000	14
15	1	Not Graduate	No	1900000	7400000	6
16	5	Not Graduate	No	4700000	10700000	10
17	2	Graduate	Yes	500000	1600000	4
18	4	Not Graduate	Yes	2900000	9400000	14
19	2	Graduate	No	2700000	10300000	10
20	5	Graduate	No	6300000	14600000	12

loan id	cibil score	residential assets_value	commercial assets_value	luxury assets_value	bank asset_value	loan status
1	778	2400000	17600000	22700000	8000000	Approved
2	417	2700000	2200000	8800000	3300000	Rejected
3	506	7100000	4500000	33300000	12800000	Rejected
4	467	18200000	3300000	23300000	7900000	Rejected
5	382	12400000	8200000	29400000	5000000	Rejected
6	319	6800000	8300000	13700000	5100000	Rejected
7	678	22500000	14800000	29200000	4300000	Approved
8	382	13200000	5700000	11800000	6000000	Rejected
9	782	1300000	800000	2800000	600000	Approved
10	388	3200000	1400000	3300000	1600000	Rejected
11	547	8100000	4700000	9500000	3100000	Approved
12	538	15300000	5800000	20400000	6400000	Rejected
13	311	6400000	9600000	14600000	4300000	Rejected
14	679	10800000	16600000	20900000	5000000	Approved
15	469	1900000	1200000	5900000	1900000	Rejected
16	794	5700000	3900000	16400000	4400000	Approved
17	663	1300000	100000	1300000	700000	Approved
18	780	2900000	2800000	6700000	4300000	Approved
19	736	1000000	0	6200000	3300000	Approved
20	652	10300000	3500000	23500000	5900000	Approved

First 20 rows. The two parts show the same rows, with different columns.

The column I want to predict is loan_status. It has two possible answers, Approved and Rejected. The other columns give information such as income, loan amount, credit score and assets. The file does not say what currency or loan-term unit is used, so I left these values as they are.

Task 3

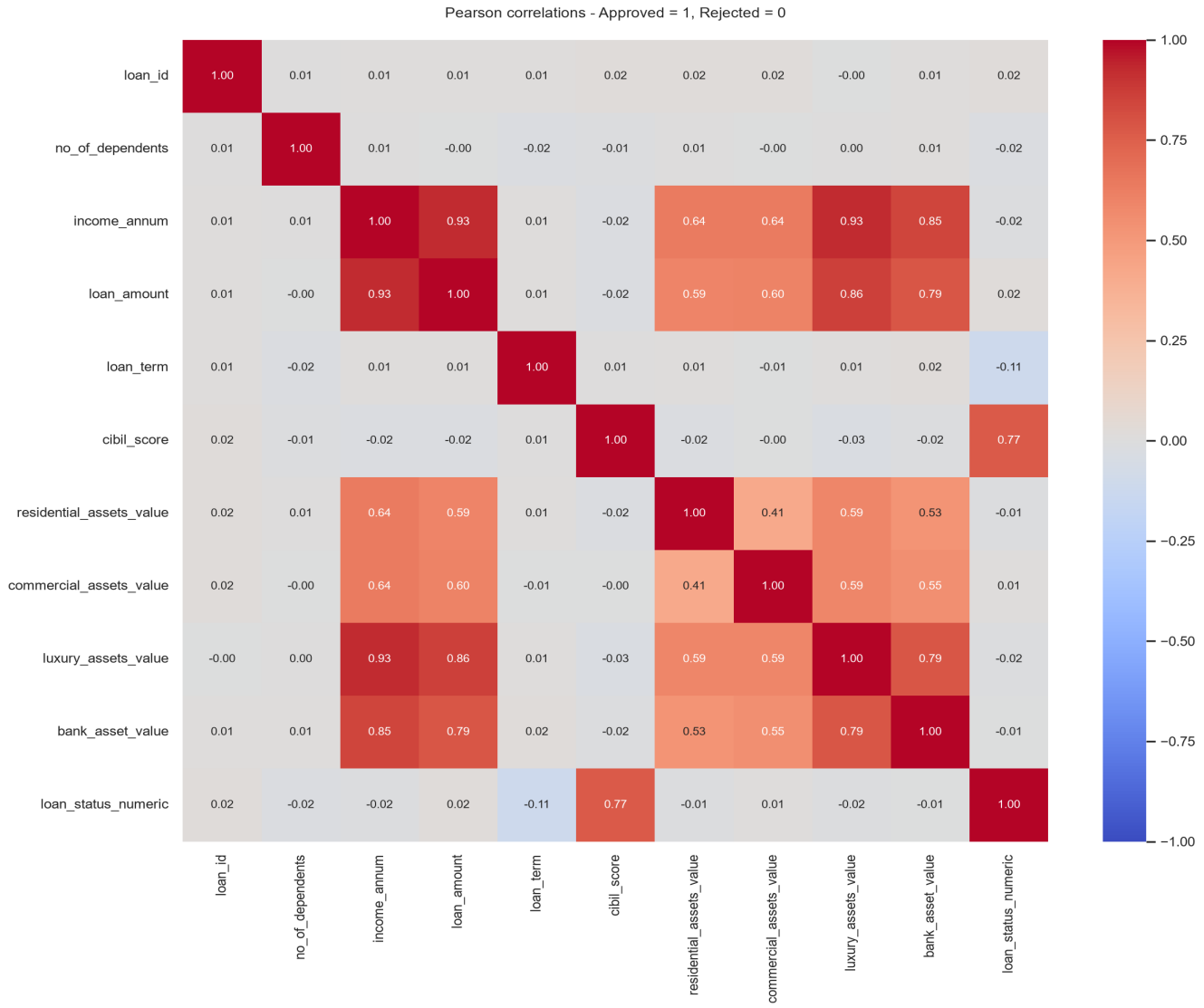
This table shows what I would expect to matter based on the meaning of each column. These are my starting opinions, rather than results calculated by Python. I included every column from the file.

Feature	Important	In-between	Not-important	Reason
loan_id			X	It is only a number used to identify the row.
no_of_dependents		X		More dependents could mean more money spent on family.
education		X		It might relate to finding a job, but does not show income directly.
self_employed		X		Income could be less steady, but this depends on the person.
income_annum	X			Someone with more income may find it easier to pay back a loan.
loan_amount	X			A larger loan means more money has to be paid back.
loan_term	X			This changes how long the person has to pay back the loan.
cibil_score	X			This gives information about how the person handled credit before.
residential_assets_value		X		Owning property could help, but it is not the same as having cash.
commercial_assets_value		X		Business assets could help, but may not be easy to sell.
luxury_assets_value		X		These show valuable belongings, but not regular income.
bank_asset_value	X			Money or assets in the bank may help with paying the loan back.
loan_status			X	This is the answer we want to predict, so it cannot be an input.

I put loan_status under Not-important because it is the answer we are trying to predict. We cannot give the program the answer as one of its inputs. loan_id is just a number for identifying each row, so I would leave that out too.

Task 4

To include loan_status in the correlation calculation, I made a new column where Approved is 1 and Rejected is 0. I used the numerical columns and calculated corrmatrix = numeric.corr(). Then I used seaborn.heatmap(corrmatrix) to make the graph. The text columns, like education, are not included in this graph.



Correlation heatmap. The values range from -1 to 1. The diagonal is 1 because each column is being compared with itself.

A value close to 1 shows that two columns tend to increase together. A value close to -1 shows that one tends to go down when the other goes up. A value near 0 shows very little straight-line relationship. Since approval is 1, a positive value with loan_status_numeric means higher values tend to go with approval.

The main thing I noticed is that cibil_score has a value of about 0.77 with approval. Income and loan amount are also strongly related to each other, but that does not mean either one is strongly related to approval. loan_id is included in the graph but is only an identifier.

Task 5

Numerical feature	Correlation	What it shows
loan_id	+0.018	Very close to zero. This is only the row identifier.
no_of_dependents	-0.018	Very small negative relationship, close to zero.
income_annum	-0.015	Very small negative relationship, close to zero.
loan_amount	+0.016	Very small positive relationship, close to zero.
loan_term	-0.113	A small negative relationship. Longer terms tend to go with rejection a bit more.
cibil_score	+0.771	Higher credit scores tend to go with approval.
residential_assets_value	-0.014	Very small negative relationship, close to zero.
commercial_assets_value	+0.008	Very small positive relationship, close to zero.
luxury_assets_value	-0.015	Very small negative relationship, close to zero.
bank_asset_value	-0.007	Very small negative relationship, close to zero.

I expected income and loan amount to matter because someone needs enough money to pay the loan back. However, their correlations with approval are very close to zero in this file. Credit score has the clearest relationship. Loan term has a smaller negative relationship. A value near zero does not prove a column is useless, because this calculation does not show every possible pattern or how several columns might work together.

Education

Education	Approved	Rejected	Total	Approval rate
Graduate	1339	805	2144	62.45%
Not Graduate	1317	808	2125	61.98%

Graduates have an approval rate of 62.45%, compared with 61.98% for non-graduates. The difference is only 0.48 percentage points. I compared percentages because the groups do not have exactly the same number of people. The rates are almost the same, so education does not seem to make much difference to approval in this dataset.

My main conclusion is that credit score would be the first column I would look at for predicting approval. These results only show relationships in this file. They do not prove that one thing causes another, and I have not tested how accurate a prediction program would be.